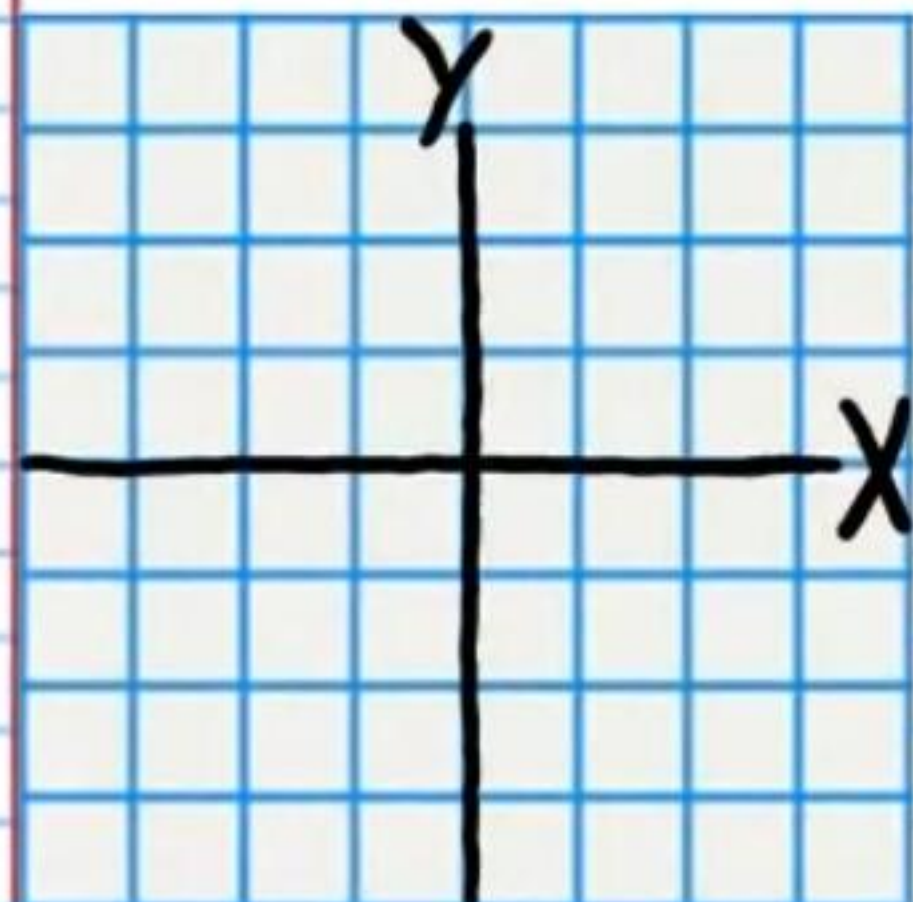


Practice Final Test-Part II (Version 2)

- I) This exam covers the second half of the course (content on Test III & Test IV).
- II) Write your name, the date, and the course title (i.e. Precalculus).
- III) Show all work.
- IV) Draw a rectangle around your final answer for each problem.
- V) A graphing calculator is required.

① Graph the following equation.

$$\theta = \frac{\pi}{4}$$



② Find the arithmetic mean of the data set below.

18, 15, 16, 20, 18

③ Bill drove his car at 68 miles per hour for 3 hours. He then drove at 60 miles per hour for 2.5 hours. What was his average speed?

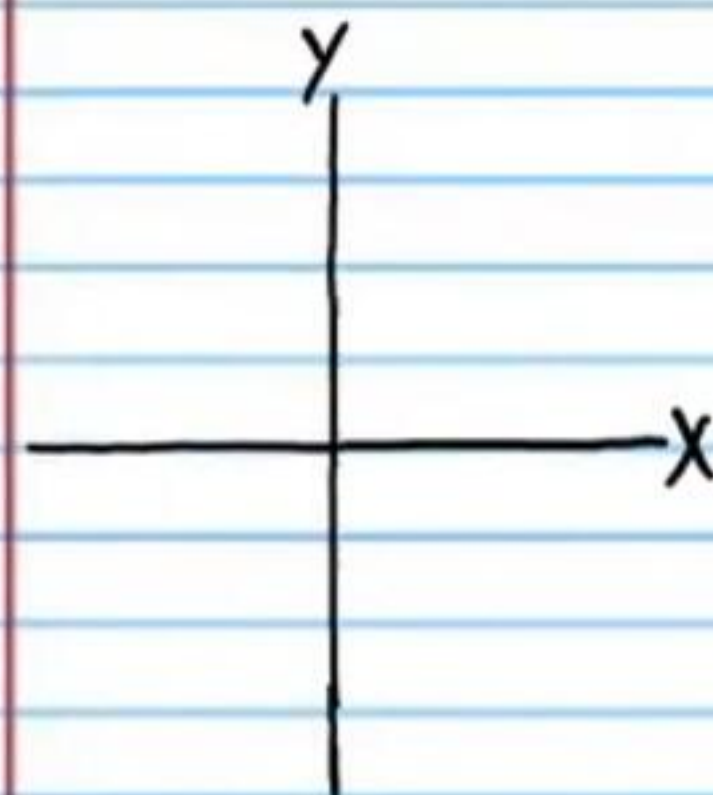
④ Convert the rectangular coordinates below to polar coordinates without a calculator. Use degree units for angle measures.

(3, -3)

Graph

Find r .

Find θ .



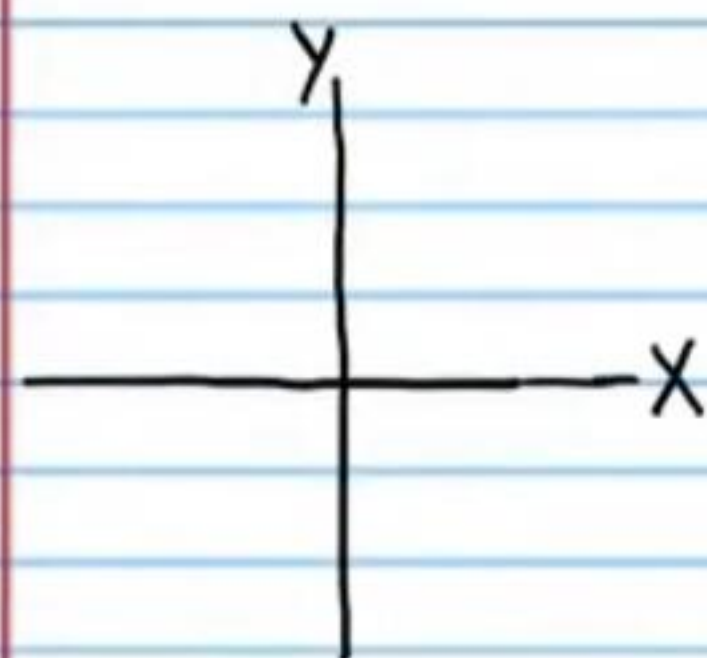
⑤ Convert the following polar coordinates to rectangular coordinates.
Round to the nearest thousandth if necessary.

$(-10, 170^\circ)$

Find x .

Find y .

Graph



⑥ Convert the equation below to rectangular form.

$$r^2 = r \sin \theta + 1$$

⑦ If $f(x) = \frac{1}{x}$, and $f(c) = 3$, find c .

⑧ Create parametric equations for x and y using t as the parameter.

$$2y - 4 = x$$

⑨ Relations are written below in various forms. Write each inverse relation in its corresponding form.

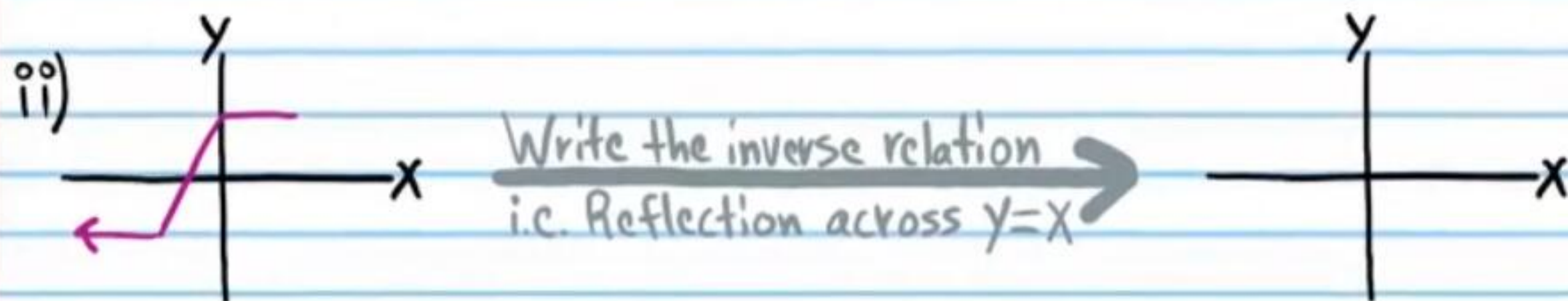
i)

X	30	20	15	19	29
Y	85	-80	-2	50	-42

 Write the inverse relation $\rightarrow$

X					
Y					

i.e. Reflection across $y=x$



iii) $x^2 + y^2 = 100$ Write the inverse relation $\rightarrow$
i.e. Reflection across $y=x$

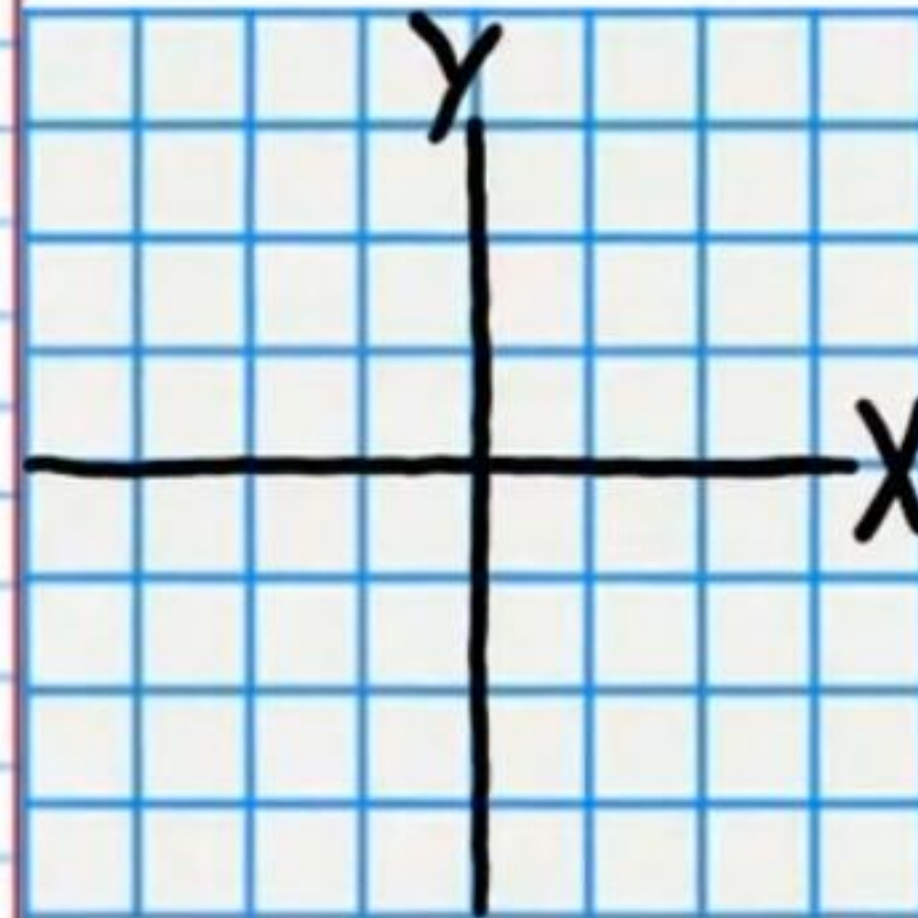
iv) $y = x^2 + \ln x$ Write the inverse relation $\rightarrow$
i.e. Reflection across $y=x$

⑩ Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot. Then eliminate the parameter.

$$x = 2\cos t, y = 2\sin t$$

Point #1

Point #2



Point #3

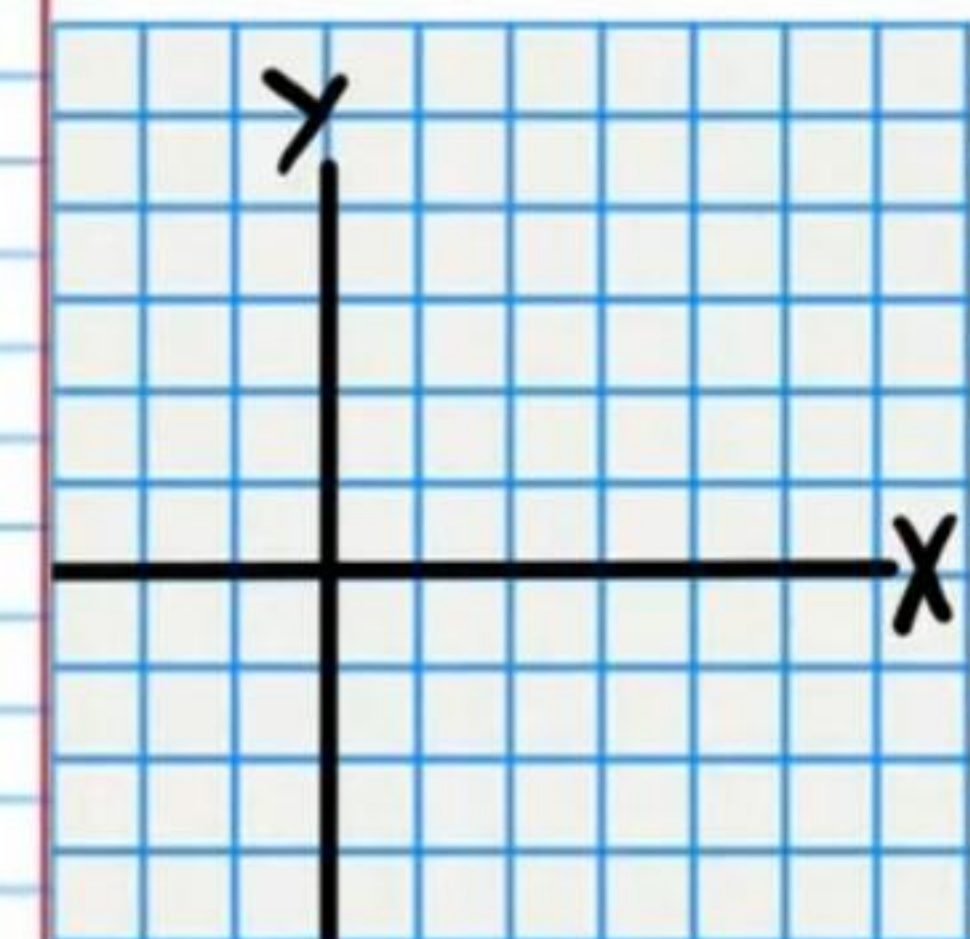
Point #4

Eliminate the parameter

$$x = 2\cos t, y = 2\sin t$$

11) Graph the following equation. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

$$r = 3 + 3\cos\theta$$



1st Point

2nd Point

3rd Point

4th Point

12) Perform the following reflections.

i)

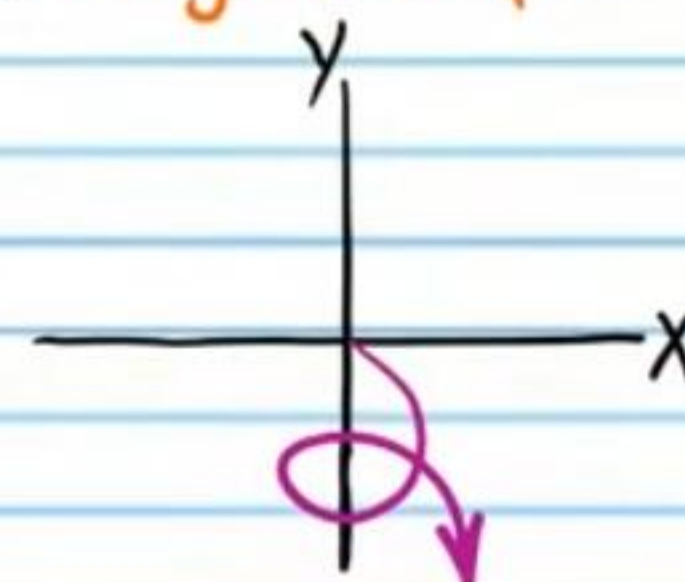
x	7	-1	10	-6	-3
y	16	-2	11	-4	0

Reflect across
the y-axis →

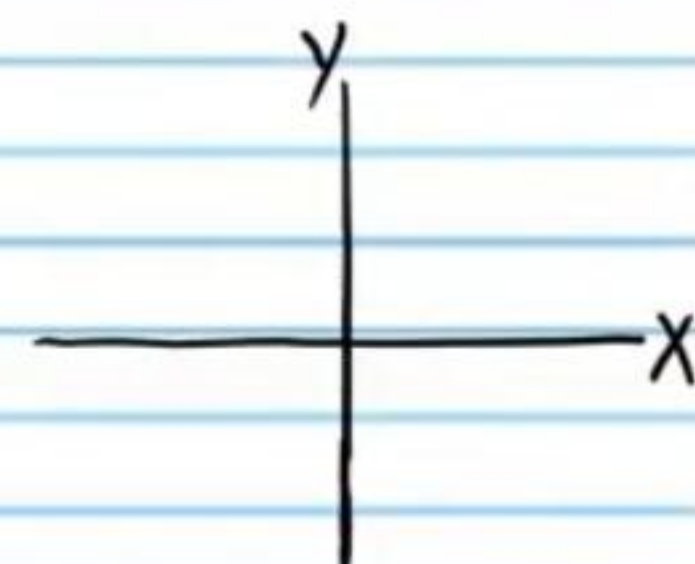
x					
y					

ii) $y = \frac{2}{x}$ Reflect across
the origin →

iii) Original Graph



Reflect across
the x-axis →



iv) $y = x^3 + 1$ Reflect across
the y-axis →

The image displays six hand-drawn graphs on lined paper, arranged in two rows of three. Each graph is plotted on a Cartesian coordinate system with x and y axes.

- Top Left:** A continuous, smooth, periodic curve resembling a sine wave, oscillating above and below the x-axis.
- Top Middle:** A discrete set of points forming a V-shape, with points in the first and second quadrants.
- Top Right:** Two horizontal line segments, one in the first quadrant and one in the third quadrant, representing a piecewise function.
- Bottom Left:** A continuous curve that forms a three-leaf rose shape, symmetric about the y-axis.
- Bottom Middle:** Two horizontal arrows, one above and one below the x-axis, pointing in opposite directions, representing a relation.
- Bottom Right:** A discrete set of points forming a horizontal line segment, with points in the first and second quadrants.

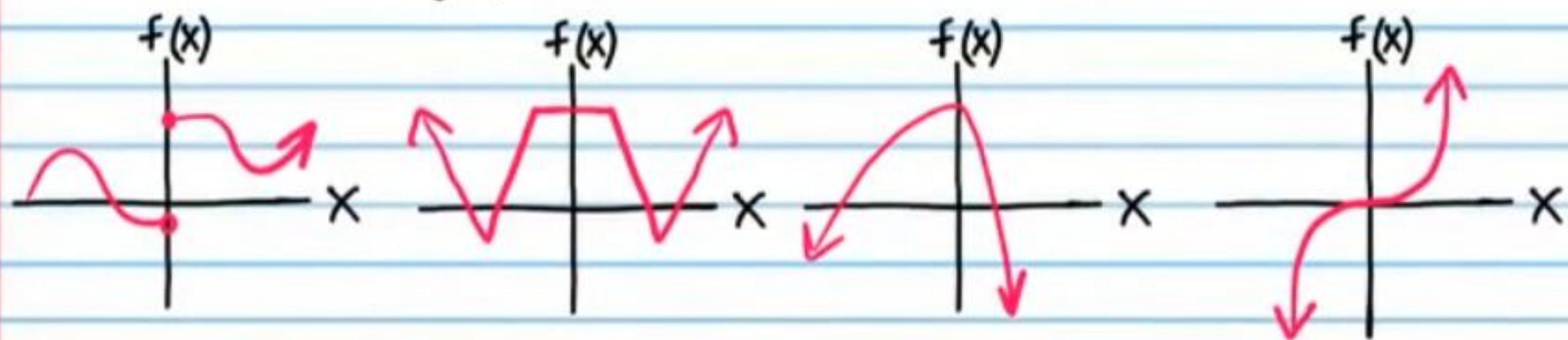
X	13	12	10	1	-2				
Y	-6	-5	-3	8	11				

$$x^2 + y^2 = \frac{1}{x^2}$$

iv) Is the graph of the equation symmetric about the origin? *Demonstrate algebraically.*

X	49	-10	20	-6	26				
f(x)	40	-4	60	-12	27				

ii) Determine if the graphs below are even functions, odd functions, or neither.



iii) Each table below represents a function "f". Write the values of the inverse function "f⁻¹". If "f" is not a one-to-one function, then you must omit values of "f⁻¹" so that it qualifies as a function. To ensure that the functions are perfectly matched inverses, values of "f" may need to be omitted also. You may choose which values to omit. Highlight or circle the omitted values.

X	2	7	16	23	4
f(x)	13	1	8	6	2

X					
f ⁻¹ (x)					

X	-6	-1	4	17	18
f(x)	-3	4	22	20	4

X					
f ⁻¹ (x)					

iv) If "f" is an odd function, and $f(-\pi) = 3$, find $f(\pi)$.

15) Determine if the function below is even, odd, or neither.

$$f(x) = x + x^2 + x^3$$

16) If $f(x) = 2(x+3)^2 - 1$ and $g(x) = 3$, find the coordinates of the point(s) where the graphs of these functions intersect.

⑪ The population of a strain of E. coli after t hours can be predicted with the equation below.

$$P(t) = \frac{2,045}{1 + 20e^{-0.21t}}$$

i) What is the carrying capacity of the strain?

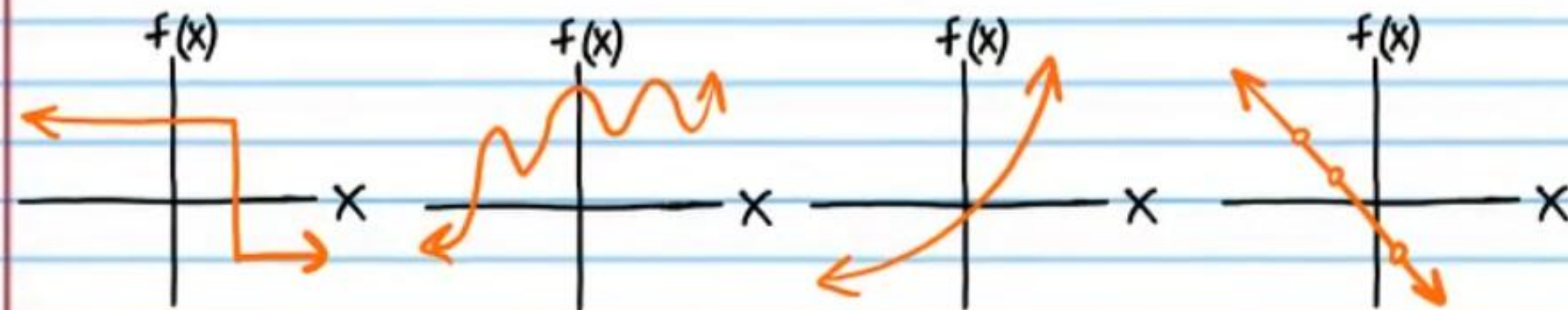
ii) What is the initial population?

iii) What will the population be after 18 hours?

iv) Use your calculator's base 10 logarithm button (LOG) and Logarithm Property IX to find the value of the logarithm below. Round to the nearest thousandth.

$$\log_{12} 25$$

⑫ i) Relations are written below in the form of graphs. Some of the graphs could represent functions that are one-to-one, some of them could represent functions that are not one-to-one, and some cannot represent functions. Use the Vertical Line Test and the Horizontal Line Test to determine the scenario for each graph. Write "1-to-1 function," "function," or "not a function."



ii) In the following problem, "f" and "g" are inverse functions. If $f(0) = 20$, find $g(20)$.

⑬ Given the following intensity of sound, find the loudness perceived by the human ear.

$$10^{-2} \text{ W/m}^2$$

②① After 15 years, a radioactive material retains 93% of its original mass. What is the half life of this radioactive material?

For problem #'s 21-22, find an equation of the inverse function.

②① $f(x) = \frac{e^x}{6} + 4$

②② $f(x) = -3\log x - 7$

②③ Show algebraically that "f" and "f⁻¹" are inverse functions.

$$f(x) = \frac{5}{x-2} \quad f^{-1}(x) = \frac{5+2x}{x}$$

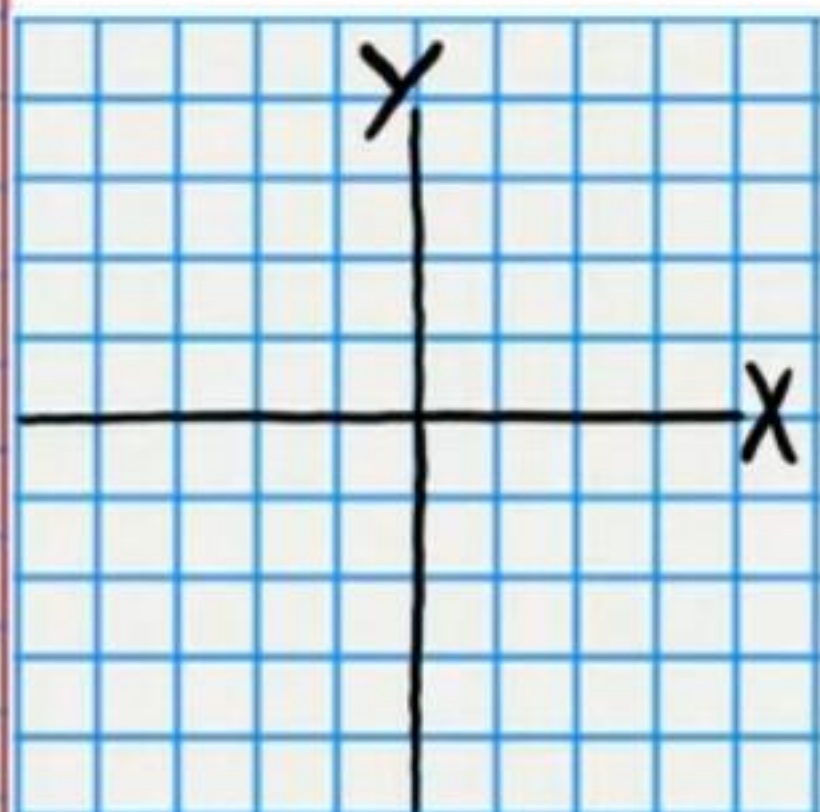
Use This Page to Show Work for Problem #23 if Necessary

②4 Graph the following equation using radian units. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

$$r = 3 \cos(2\theta)$$

1st Point

2nd Point

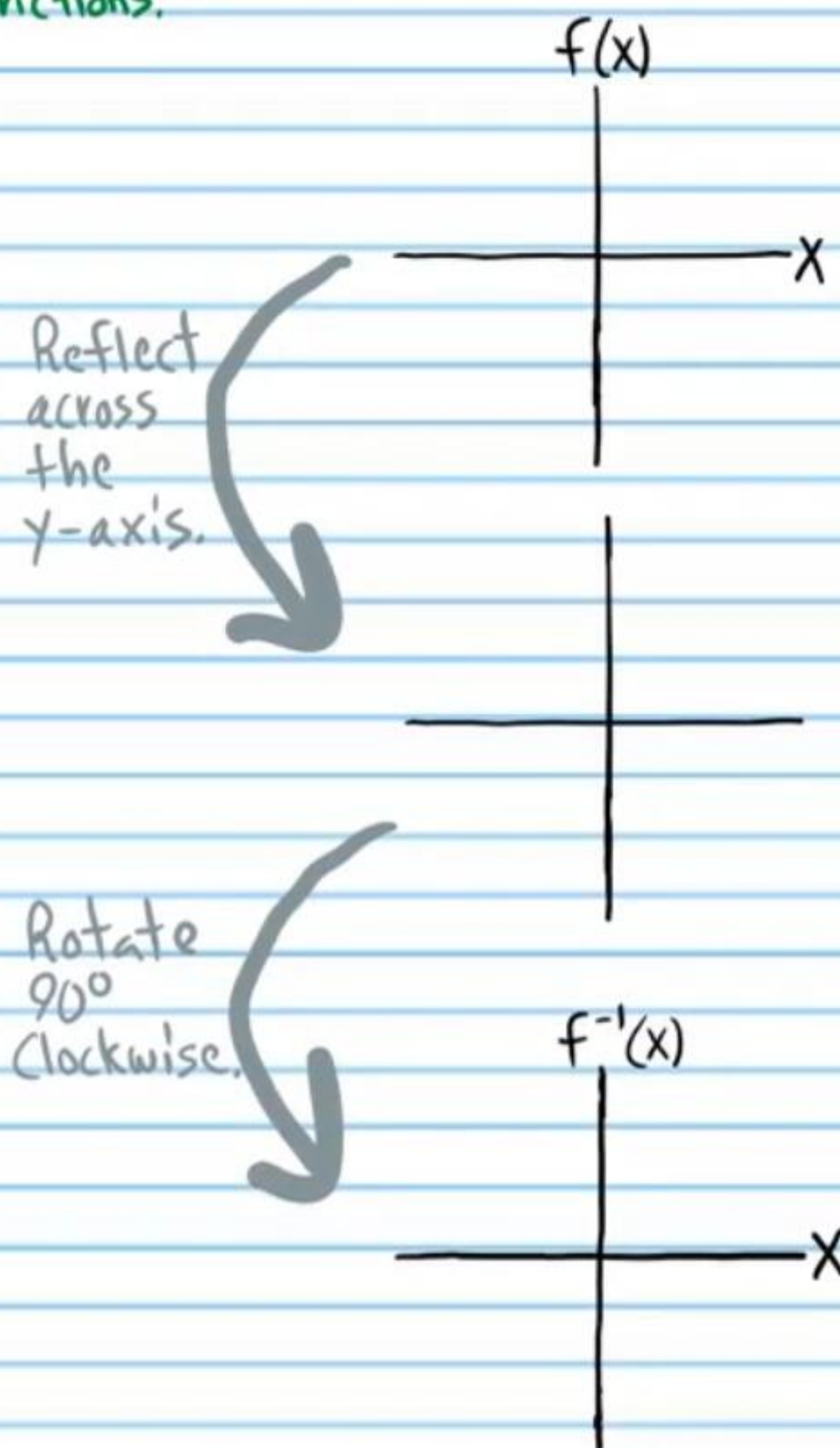


3rd Point

4th Point

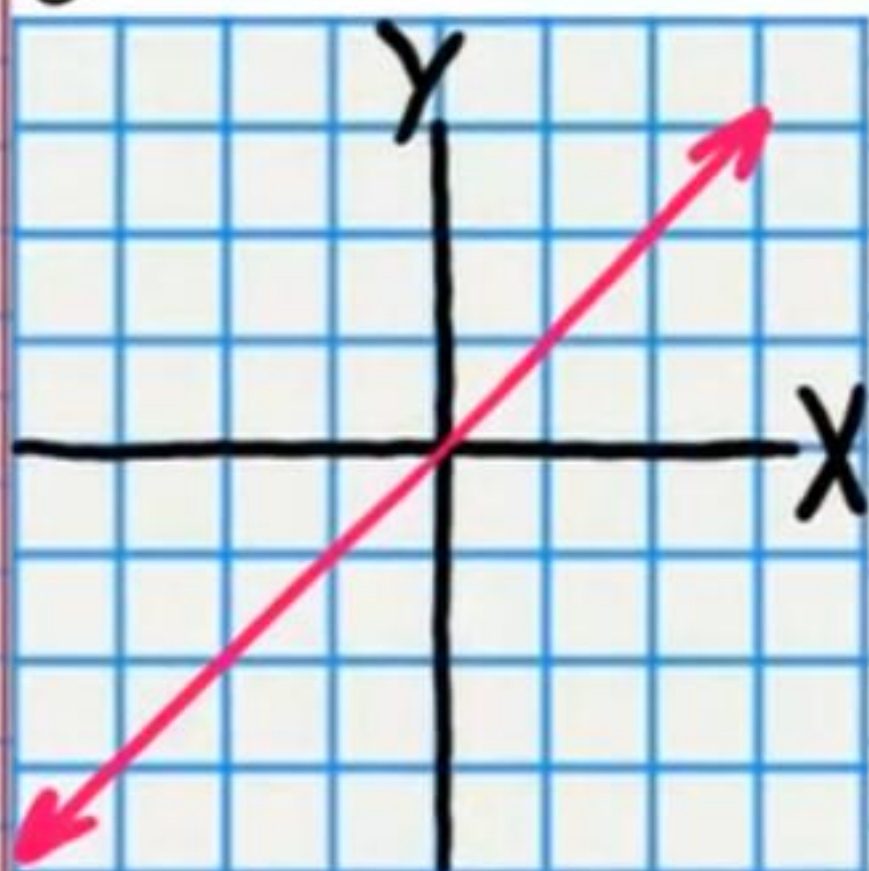
②5 Find the equation of the inverse function. Restrictions must be used on the original function and/or inverse function so that they are perfectly matched and no aberrations occur. To understand this, graph the function and note that it fails the Horizontal Line Test, which means that it is not one-to-one. Draw the inverse graph and note that it fails the Vertical Line Test. Highlight the parts of the inverse graph and the original graph that you will omit and use inequality notation next to the equations to indicate restrictions.

$$f(x) = \csc x$$



Answers

①



② 17.4

③ 64.36 m/h

④ $(3\sqrt{2}, -45^\circ)$ or $(3\sqrt{2}, 315^\circ)$ or $(-3\sqrt{2}, 135^\circ)$ * Other variations are possible.

⑤ $(9.848, -1.736)$

⑥ $x^2 + y^2 = y + 1$

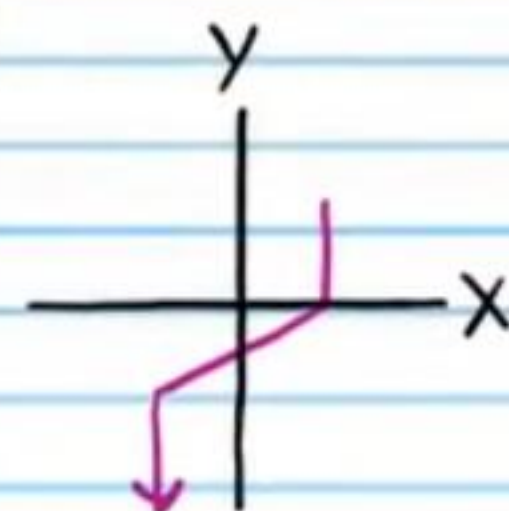
⑦ $c = \frac{1}{3}$

⑧ $y = t, x = 2t - 4$ or $x = t, y = \frac{t+4}{2}$

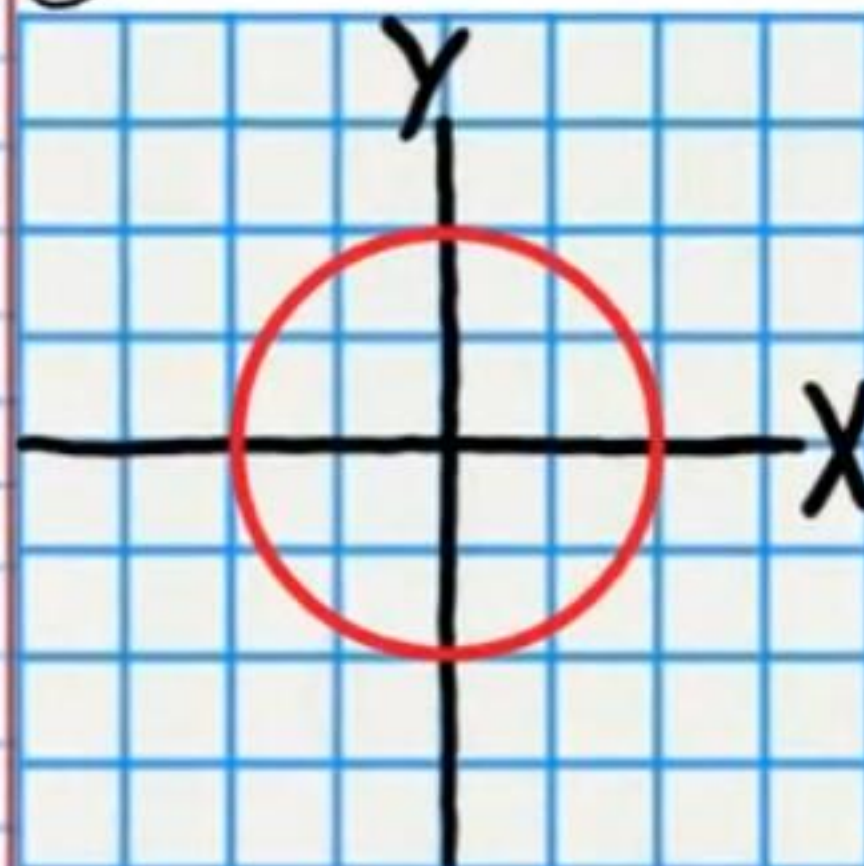
⑨ i)

x	85	-80	-2	50	-42
y	30	20	15	19	29

ii)



⑩

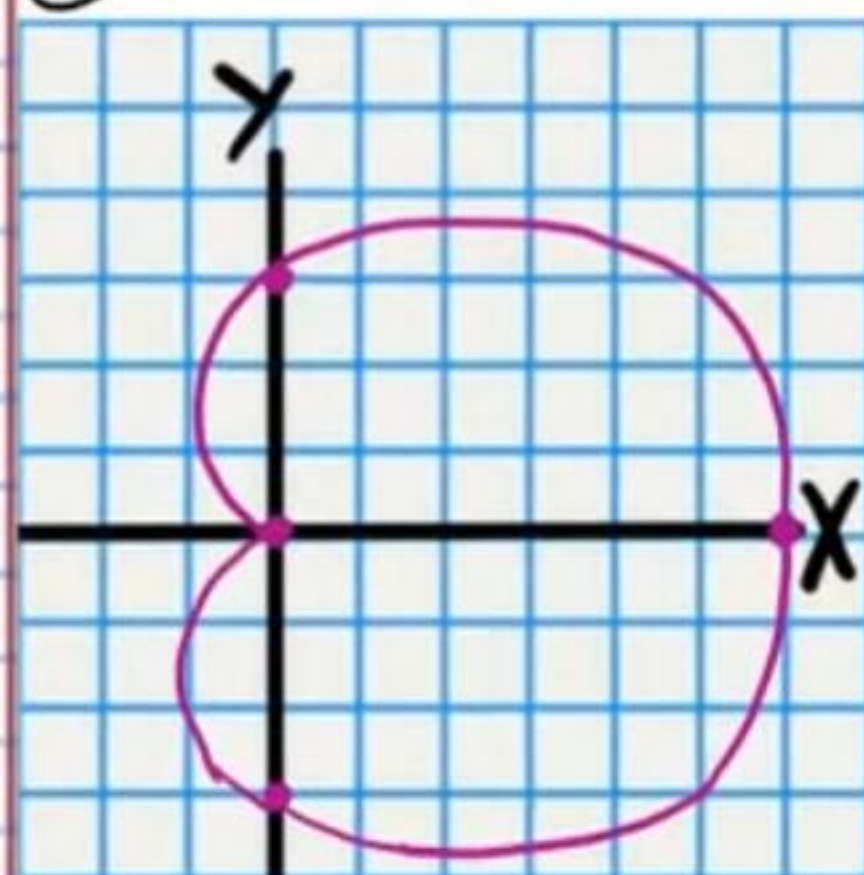


Other points are possible. Only 4 are necessary.

$(0,2) (2,0) (-2,0) (0,-2)$

$x^2 + y^2 = 4$

⑪



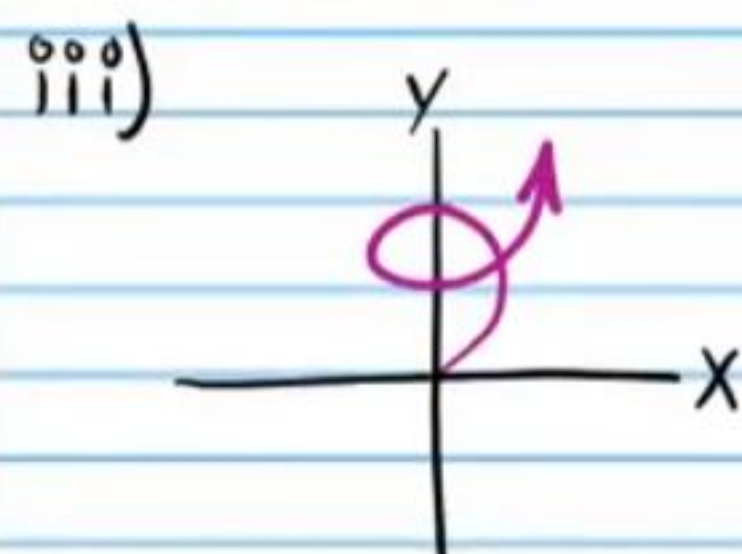
Other points are possible. Only 4 are necessary.

$(6,0) (3,90^\circ) (0,180^\circ) (3,270^\circ)$
 $(4.5, 60^\circ) (4.5, -60^\circ)$

12 i)

x	-7	1	-10	6	3
y	16	-2	11	-4	0

ii) $y = \frac{2}{x}$



iv) $y = -x^3 + 1$

13 i) Y O N
X XYO XYO

ii)

X	13	12	10	1	-2	-13	-12	-10	-1	2
Y	-6	-5	-3	8	11	-6	-5	-3	8	11

iii) $x^2 + (-y)^2 = \frac{1}{x^2}$ iv) $(-x)^2 + (-y)^2 = \frac{1}{(-x)^2}$

$$x^2 + y^2 = \frac{1}{x^2}$$

Yes

$$x^2 + y^2 = \frac{1}{x^2}$$

Yes

14 i)

X	49	-10	20	-6	26	-49	10	-20	6	-26
f(x)	40	-4	60	-12	27	40	-4	60	-12	27

ii) Neither Even Neither Odd

iii)

X	2	7	16	23	4
f(x)	13	1	8	6	2

X	13	1	8	6	2
f ⁻¹ (x)	2	7	16	23	4

X	-6	-1	4	17	18
f(x)	-3	4	22	20	4

X	-3	4	22	20	4
f ⁻¹ (x)	-6	-1	4	17	18

Omit the yellow points or the blue points.

iv) $f(\pi) = -3$

15 $f(-x) = -x + (-x)^2 + (-x)^3$

$$f(-x) = -x + x^2 - x^3$$

Neither

16 $(\sqrt{2} - 3, 3)$ $(-\sqrt{2} - 3, 3)$

17 i) 2,045 E. coli

ii) 97 E. coli

iii) 1,404 E. coli

iv) 1.295

(18) i) Not a Function Function 1-to-1 Function 1-to-1 Function

ii) $g(20) = \boxed{0}$

(19) $\boxed{100 \text{ dB}}$

(20) $\boxed{143.27 \text{ years}}$

(21) $f^{-1}(x) = \ln(6x - 24)$

(22) $f^{-1}(x) = 10^{\frac{x+7}{-3}}$

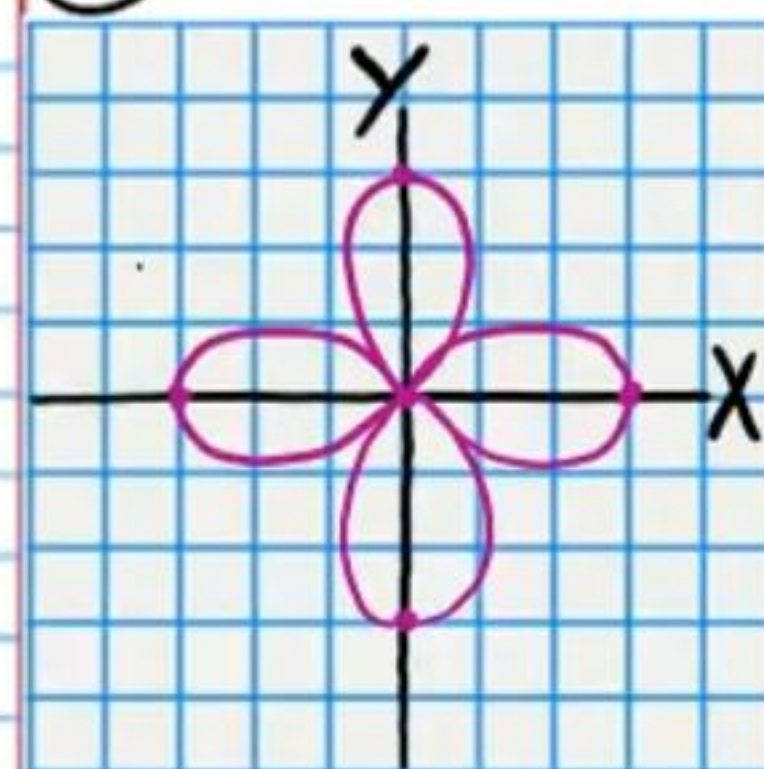
(23) $f(f^{-1}(x)) = \frac{5}{\frac{5+2x}{x} - 2} = \frac{5}{\frac{5+2x}{x} - \frac{2x}{x}} = \frac{5}{\frac{5}{x}} = x \checkmark$

$$f^{-1}(f(x)) = \frac{5 + 2\left(\frac{5}{x-2}\right)}{\frac{5}{x-2}} = \frac{5 + \frac{10}{x-2}}{\frac{5}{x-2}} = \frac{\frac{5x-10}{x-2} + \frac{10}{x-2}}{\frac{5}{x-2}} = \frac{\frac{5x}{x-2}}{\frac{5}{x-2}} = \frac{5x}{x-2} \div \frac{5}{x-2}$$

$$\frac{5x}{x-2} \cdot \frac{x-2}{5} = x \checkmark$$

$\boxed{\text{Yes}}$

(24)



Other points are possible. Only 4 are necessary.

$$\left(0, \frac{\pi}{4}\right) \quad (3, 0) \quad \left(-3, \frac{\pi}{2}\right) \quad \left(-3, -\frac{\pi}{2}\right)$$

$$\left(\frac{3}{2}, -\frac{\pi}{6}\right) \quad \left(\frac{3}{2}, \frac{\pi}{6}\right) \quad \left(-\frac{3}{2}, \frac{2\pi}{3}\right)$$

(25) $f(x) = \csc x, \quad -90^\circ \leq x \leq 90^\circ, \quad x \neq 0^\circ$
 $f^{-1}(x) = \csc^{-1} x, \quad -90^\circ \leq y \leq 90^\circ, \quad y \neq 0^\circ$

